

Technical Specification: Automated Fluid Dispensing System

1. System Overview

This document outlines the engineering architecture of an automated, closed-loop fluid dispensing system. The mechatronic assembly integrates optoelectronic proximity sensing, solid-state power switching, and a mechanical centrifugal pump to achieve touchless operation. The system acts as a transducer, converting external spatial perturbations into electrical signals, which are subsequently amplified to produce hydraulic work.

2. Optoelectronic Proximity Detection

The detection subsystem relies on active infrared (IR) reflectance principles. An IR light-emitting diode (LED) emits photons at a specific wavelength (typically $\lambda \approx 940$ nm).

2.1 Physics of Reflectance

When an object (e.g., a human hand) enters the operational envelope, it scatters the incident infrared radiation. The intensity of the reflected light, I_r , follows the inverse-square law combined with the Lambertian reflectance of the surface (R_d):

$$I_r \propto \frac{I_0 \cdot R_d}{d^2}$$

where I_0 is the source intensity and d is the distance to the target.

2.2 Signal Conditioning

The reflected photons are captured by a matched phototransistor, generating a photocurrent (I_p) proportional to the irradiance. To digitize this analog phenomenon, the signal is fed into a precision voltage comparator (e.g., LM393). To prevent "chattering" (rapid switching of the pump at the threshold boundary), the comparator circuit incorporates Schmitt trigger hysteresis. The output state V_{out} transitions only when the input voltage V_{in} crosses designated upper (V_{TH}) or lower (V_{TL}) thresholds:

$$V_{TH} = V_{ref} + \Delta V, \quad V_{TL} = V_{ref} - \Delta V$$

3. Motor Activation, Power Electronics, and EMI

The logic-level output from the comparator or Microcontroller Unit (MCU) (e.g., 3.3V or 5V, < 20 mA) is insufficient to drive a DC motor, which represents a highly inductive, high-current load.

3.1 Solid-State Switching and Thermal Management

A logic-level N-channel Power MOSFET (e.g., IRLZ44N) is utilized as a high-speed electronic switch. Operating in the saturation region, the MOSFET connects the motor to ground when the

Gate-to-Source voltage (V_{GS}) exceeds the threshold voltage (V_{th}). The power dissipation (P_{loss}) in the switch comprises conduction and switching losses:

$$P_{loss} = (I_D^2 \cdot R_{DS(on)}) + \left(\frac{1}{2} V_{DS} \cdot I_D \cdot (t_{on} + t_{off}) \cdot f_{sw} \right)$$

Thermal management is critical to prevent junction breakdown. The steady-state junction temperature (T_J) must remain below the absolute maximum rating (175°C typically), dictated by the ambient temperature (T_A) and the thermal resistance from junction-to-ambient (θ_{JA}):

$$T_J = T_A + P_{loss} \cdot \theta_{JA}$$

3.2 Inductive Flyback Protection

A DC motor acts as a large inductor (L). When the MOSFET abruptly terminates the current ($di/dt < 0$), Faraday's Law of Induction dictates a massive reverse voltage spike:

$$V_{emf} = -L \frac{di}{dt}$$

To protect the MOSFET from avalanche breakdown, a flyback diode (e.g., 1N4007 Schottky) is placed in parallel with the motor in a reverse-biased configuration, providing a safe discharge path for the stored magnetic energy ($E = \frac{1}{2} LI^2$).

3.3 Electromagnetic Interference (EMI) Suppression

Brushed DC motors generate significant wideband EMI due to the mechanical commutation (arcing at the brushes). High dv/dt transients can couple back into the MCU logic, causing spontaneous resets. To mitigate conducted and radiated emissions, three multi-layer ceramic capacitors (MLCCs, typically 100nF) are deployed: one bridging the motor terminals, and two bypassing each terminal to the metallic motor chassis (Earth ground).

4. Control Systems Integration

If variable flow rate regulation is required, an MCU (e.g., ATmega328P or STM32) driving the MOSFET via Pulse Width Modulation (PWM) is introduced.

4.1 System Transfer Function

The DC motor and pump assembly can be modeled mathematically in the Laplace domain. The transfer function relating the input voltage $V(s)$ to the angular velocity $\omega(s)$ is a second-order linear time-invariant (LTI) system:

$$G(s) = \frac{\omega(s)}{V(s)} = \frac{K_t}{(L_a s + R_a)(J s + b) + K_t K_e}$$

where:

- L_a, R_a = Armature inductance and resistance.
- J, b = Rotor inertia and viscous friction coefficient.
- K_t, K_e = Motor torque and back-EMF constants.

4.2 PID Flow Control

By integrating a Hall-effect flow sensor into the hydraulic loop, a closed-loop discrete Proportional-Integral-Derivative (PID) controller can be implemented to maintain a precise volumetric flow rate despite varying head pressures:

$$u[k] = K_p e[k] + K_i \sum_{j=0}^k e[j] \Delta t + K_d \frac{e[k] - e[k-1]}{\Delta t}$$

5. Fluid Dynamics and Water Dispensing

The electromechanical conversion utilizes a micro-centrifugal pump to generate hydraulic power.

5.1 Hydraulic Power and the Navier-Stokes Context

The impeller's rotation imparts kinetic energy to the fluid, translating into an increase in static pressure (Δp) and flow rate (Q). Assuming incompressible, viscous flow, the hydraulic power (P_{hyd}) delivered to the fluid is given by:

$$P_{hyd} = \rho \cdot g \cdot Q \cdot H_{total}$$

The total dynamic head (H_{total}) must account for major friction losses in the piping (h_f), calculated via the Darcy-Weisbach equation:

$$h_f = f_D \frac{L}{D} \frac{v^2}{2g}$$

5.2 Cavitation and Net Positive Suction Head (NPSH)

To ensure the pump operates without destructive vapor bubble collapse (cavitation) on the impeller blades, the Net Positive Suction Head Available ($NPSH_A$) must strictly exceed the pump's Required NPSH ($NPSH_R$):

$$NPSH_A = \frac{P_{abs} - P_v}{\rho g} - h_z - h_f \geq NPSH_R$$

where P_{abs} is the absolute pressure at the fluid surface, P_v is the vapor pressure of the fluid at the operating temperature, and h_z is the static suction lift.

6. Components (Bill of Materials)

For a robust, university-grade implementation, the following components are required:

1. **Sensor:** TCRT5000 IR Reflective Sensor Module (incorporating IR LED, phototransistor, and LM393 comparator with adjustable potentiometer).
2. **Actuator:** 5V-12V DC Submersible Centrifugal Pump (Brushless DC preferred for longevity, zero brush EMI, and higher MTBF).
3. **Power Switching:** IRLZ44N Logic-Level N-Channel Power MOSFET (low $R_{DS(on)}$ for reduced thermal dissipation).
4. **Protection & Filtering:** 1N5819 Schottky Flyback Diode; $3 \times 100\text{nF}$ Ceramic Capacitors.
5. **Control Logic:** Microcontroller (e.g., STM32F103 or ESP32) for implementing 20kHz+ PWM (supersonic to avoid audible magnetostriction noise) and digital PID debounce algorithms.
6. **Power Supply:** A regulated DC Switch Mode Power Supply (SMPS) capable of handling motor stall currents (e.g., 12V/3A).
7. **Hydraulics:** Silicone or PVC tubing mathematically sized to match the pump's discharge nozzle to minimize Reynolds number (Re) turbulence losses.

7. Advanced Applications

Beyond simple residential utility, the principles of this system scale to advanced engineering applications:

- **Precision Agriculture:** Integration with capacitive soil moisture sensors and an MCU to create an automated evapotranspiration-based irrigation matrix, optimizing water expenditure using PID control loops.
- **Aseptic Medical Dispensing:** Deployment in surgical scrubbing stations where cross-contamination vectors must be mathematically reduced to near-zero probabilities through strict non-contact actuation.
- **Industrial Chemical Dosing:** Utilizing peristaltic pump variants instead of centrifugal pumps for the highly accurate, time-dependent volumetric dosing of corrosive fluids or reagents, factoring in non-Newtonian fluid behaviors.